

Ajay Singh

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Education

Georgia Institute of Technology

Atlanta, Georgia

B.S. IN COMPUTER SCIENCE | MAJOR GPA: 3.5 | DEAN'S LIST

Aug. 2013 - May 2020

- Concentrations: Artificial Intelligence, Security, Information Management

Skills

Languages: Golang, Python, TypeScript, Bash, SQL

Infrastructure: Kubernetes, OpenShift, Terraform, ArgoCD, GCP, Linux

Tools: Git, Tekton, Splunk, Ansible

Work Experience

Ford Motor Company

Dearborn, MI

KUBERNETES PLATFORM ENGINEER

Jan. 2023 - Present

- Architected and operated a multi-cluster Kubernetes/OpenShift platform supporting 3,000+ developers across 100+ hybrid (on-prem + multi-cloud) clusters, owning reliability, capacity planning, and upgrade strategy.
- Engineered GitOps-based platform automation using ArgoCD and Terraform to standardize namespace provisioning, policy enforcement, and application delivery, reducing manual operations and improving onboarding speed.
- Implemented policy enforcement via admission controllers reducing misconfigured deployments by 42%.
- Defined multi-tenant namespace isolation patterns, quota standards, and platform guardrails that improved cluster utilization and reduced noisy-neighbor risk across shared environments.
- Built and operated internal platform APIs and automation services with CI/CD, observability, and production SLOs, improving deployment reliability and reducing operator toil.
- Hardened Terraform modules with version pinning, policy validation, and secure defaults, reducing configuration drift and enabling safer infrastructure rollouts at scale.
- Served in primary and escalation on-call rotations for a 24/7 production Kubernetes platform, leading incident triage, root cause analysis, and systemic remediation to reduce repeat failure modes.
- Designed and executed phased OpenShift and control-plane upgrade strategy across 100+ clusters, maintaining N-1 compliance and zero customer downtime.
- Drove platform adoption and operational consistency by leading recurring Kubernetes forums focused on architecture changes, reliability practices, and developer enablement.

DATA ENGINEER

Jan. 2022 - Dec. 2022

- Designed and maintained distributed data pipelines supporting connected vehicle services, handling ingestion, transformation, and storage at production scale.
- Implemented monitoring and alerting tooling to improve data pipeline observability and ensure high availability of downstream analytics systems.
- Delivered data infrastructure supporting sales, service, and warranty analytics for 384,000 vehicles and 79,000 EU customers.
- Led migration of COBOL-based mainframe data pipelines to distributed system, eliminating \$800K annual infrastructure cost and improving maintainability.

SOFTWARE DEVELOPER

Sep. 2020 - Dec. 2021

- Developed internal enterprise application used to reconcile employee smart services across campus environments.
- Optimized performance and reliability of COVID screening platform serving 80,000+ daily users.
- Reduced SMS infrastructure spend by 28% through architectural optimization and vendor-efficiency improvements.

PRODUCTION SUPERVISOR

Jun. 2020 - Sep. 2020

- Supported ventilator production in partnership with GE Healthcare and 3M during a critical manufacturing ramp for FEMA.
- Managed multiple shifts for the entire manual calibration and testing process across 9 separate critical quality checks resulting in a 91.5% FTT.
- Led operations across multiple shifts and quality checkpoints, improving throughput and team execution in a high-pressure environment.

Mailchimp

Atlanta, GA

SOFTWARE ENGINEER INTERN | NETWORKS

Aug. 2017 - Aug. 2019

- Built incident management tool to coordinate network events and streamline cross-team communication.
- Automated network change workflows using Ansible + GitHub, reducing configuration errors by 27% across 300+ devices.
- Developed a Slackbot and REST-based automation to alert engineers on configuration changes and privileged-access events.
- Performed security audits and traffic analysis across 200+ network devices to improve stability and reduce peak bandwidth consumption.

Georgia Tech Cyber Security Department

Atlanta, GA

JUNIOR SECURITY ANALYST

May 2017 - Dec. 2018

- Monitored security alerts and triaged over 300 security incidents.
- Developed tools in Python to remove compromised users from currently logged in services.
- Researched potential security threats and updated alerts for the entire Georgia Tech Network.
- Identified compromised systems and followed up appropriately with department systems administrators.
- Contributed to the effort to automate many of the established security processes.